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DSC 640: Data Presentation & Visualization

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National Database of Childcare Prices

I analyzed the National Database of Childcare Prices covering the years 2008 through 2018. This dataset includes county and state-level measures of childcare costs, household income, unemployment, and labor force participation. To begin, I cleaned the dataset by removing variables that only flagged imputed values, since I chose to use the imputed values directly. My initial exploratory analysis focused on national and state trends in childcare costs, female labor force participation, unemployment rates, and affordability.

The first finding was that childcare costs have risen steadily over the decade. Center-based infant care climbed from \$133 per week in 2008 to more than \$180 per week in 2018, while family-based care also increased, though at a lower level. At the same time, both male and female unemployment rates spiked in 2008-2009 and then declined through 2018, which means families were more often employed yet also facing higher childcare costs. Female labor force participation among mothers and children under six remained steady near 70%, suggesting that even as costs rose, families continued to rely on dual incomes to sustain themselves. When I calculated affordability directly by comparing annualized childcare costs to median household income, the results were surprising. On

average, families spent about 16% to 17% of their income on center-based infant care and about 12% to 13% on family-based care. The federal benchmark for affordable childcare, defined by the Department of Health and Human Services, is 7% of income. This means that families consistently spend two to three times more than what is considered affordable.

While analyzing the data, I assumed that weekly childcare costs should be annualized to compare properly against annual household income. I also excluded missing values when averaging by state or year, and I treated county and state-level median income as representative of affordability in that region. Some things remain unclear, such as how costs vary within counties across racial or socioeconomic groups, and to what extent childcare availability, not just price, influences labor force participation. Another open question is the effectiveness of government assistance or tax credits in reducing the burden on families.

The direction of my story will be on how childcare in the U.S. is unaffordable. Costs have increased for more than a decade, while labor force participation has not shifted significantly, and families are forced to spend far more than the federal affordability benchmark. This suggests that without meaningful policy intervention, childcare will remain a big burden that limits both household wellbeing and economic mobility. The message I want to communicate is that childcare affordability is not a discretionary issue; it is an essential public policy challenge that requires governmental action.

The target audience for this project is policymakers. Legislators, government agencies, and advisory boards are the decision-makers who can address childcare affordability through government assistance, tax credits, or direct investment in childcare infrastructure. To reach this audience, I will use 3 different mediums. The first is a billboard, which will reach the general public and build political pressure by displaying a simple but powerful message: “Families are paying more than double what childcare should cost.” The second is a PowerPoint presentation that turns the data into a clear story with visuals, designed for use in legislative hearings or stakeholder meetings. The third is an interactive dashboard, which will allow policymakers and their staff to explore trends by state and year, compare costs to household incomes, and directly observe where the affordability crisis is most severe. Together, these 3 mediums will ensure that the message reaches both the public and decision-makers in forms that are impactful, accessible, and persuasive.

My design approach focused on accessibility, clarity, and audience impact. In Power BI, I used consistent color coding to help policymakers easily interpret differences across states and years. The interactive filters I added allow the audience to explore patterns over time, while my map visualization utilizes color gradients that accurately reflect affordability levels. In the PowerPoint presentation, I emphasized visual storytelling, readable fonts, and balanced spacing, ensuring that each visual supported the main message without overwhelming the audience. The billboard design followed a minimalist style with strong contrast and icons to convey the cost imbalance in just a few seconds of viewing time. I improved the billboard from my previous design by adding a simple bar chart comparing the average annual childcare cost to the 7% federal affordability benchmark. This visual

addition made the message more data-driven while preserving the minimalist style. I chose bold colors and labeled dollar values directly on the bars to ensure clarity for viewers. The billboard allows the general public to instantly grasp how much more families are paying compared to what is considered affordable. Across all mediums, I avoided clutter, ensured titles summarized insights, and made visuals readable for both technical and general audiences.

Since this project uses government data, ethical use centered on accuracy, transparency, and fairness in representation. I ensured all visualizations began at zero to avoid distortion, included clear labeling for cost units and time periods, and avoided implying causal relationships that the data could not fully support. The intent was to raise awareness and promote informed policy discussion. This project highlighted the importance of aligning data design choices with audience needs. It showed how simplifying visuals without oversimplifying meaning can create a strong impact. I learned to use Power BI not just for visualization but as a tool for storytelling that makes data interactive.

If I were to do this project again, I would broaden the analysis to include childcare availability and demographic factors, including race, to better identify which populations are most impacted by affordability disparities. I would also experiment with time-series forecasting to visualize potential future affordability under different policy scenarios. I enjoyed transforming raw data into something meaningful for real-world audiences. I learned how to make data interactive by using Power BI, which made my data become “alive” and made it more interesting to work with.

References

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